PHASE 4: Applied Data Analytics Capstone Project

Introduction:

The purpose of this dashboard is to analyze e-commerce sales performance and identify key factors influencing customer conversion, retention, revenue generation, and customer satisfaction. By integrating data from customers, orders, products, reviews, and geographic locations, the dashboard provides a comprehensive view of business performance across different dimensions. The analysis focuses on understanding customer purchasing behavior, product category performance, regional sales trends, delivery efficiency, and customer feedback. These insights help businesses make data-driven decisions to improve operational efficiency, enhance customer experience, increase repeat purchases, and drive sustainable revenue growth.

Business Questions:

KPIs:

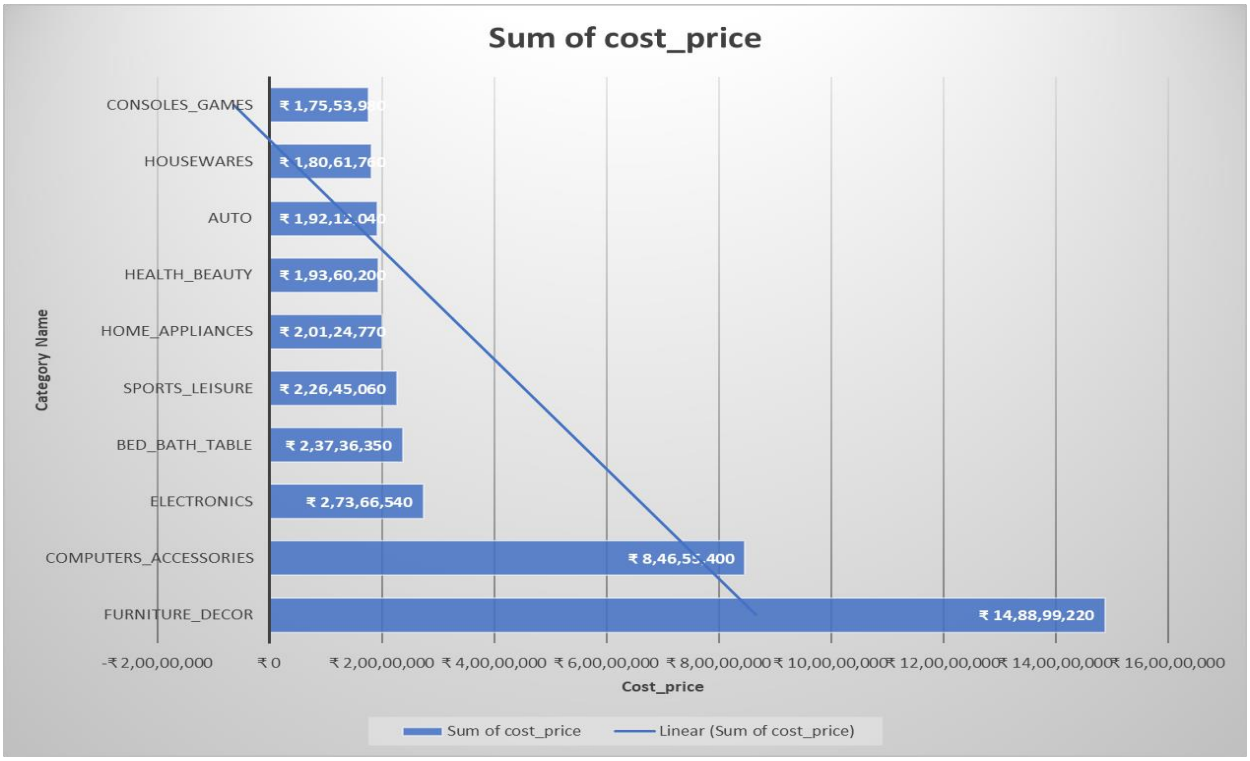
1.What is the AOV across discount bands and customer segments?
₹. 8,626.95
2.What is the return rate across customer segments?
0.99961
3.What is the on-time delivery rate?
0.9985
4.What is the average delivery delay?
40.9282
5. What is the average review score?
4.0908

Main Questions:

1. Which Product Categories Generate the Top 10 Highest Revenue?
2. Which Categories Have the Top 10 Highest Order Volume?
3. What is the Repeat Purchase Rate?
4. How Does Delivery Performance Affect Customer Satisfaction?
5. Which Product Categories Receive the Best Reviews?
6. How do sales revenues vary across different months?
7. How has overall sales performance changed over the years?
8. How does product pricing influence customer satisfaction and review ratings?

1. Which Product Categories Generate the Top 10 Highest Revenue?

Pivot Chart:

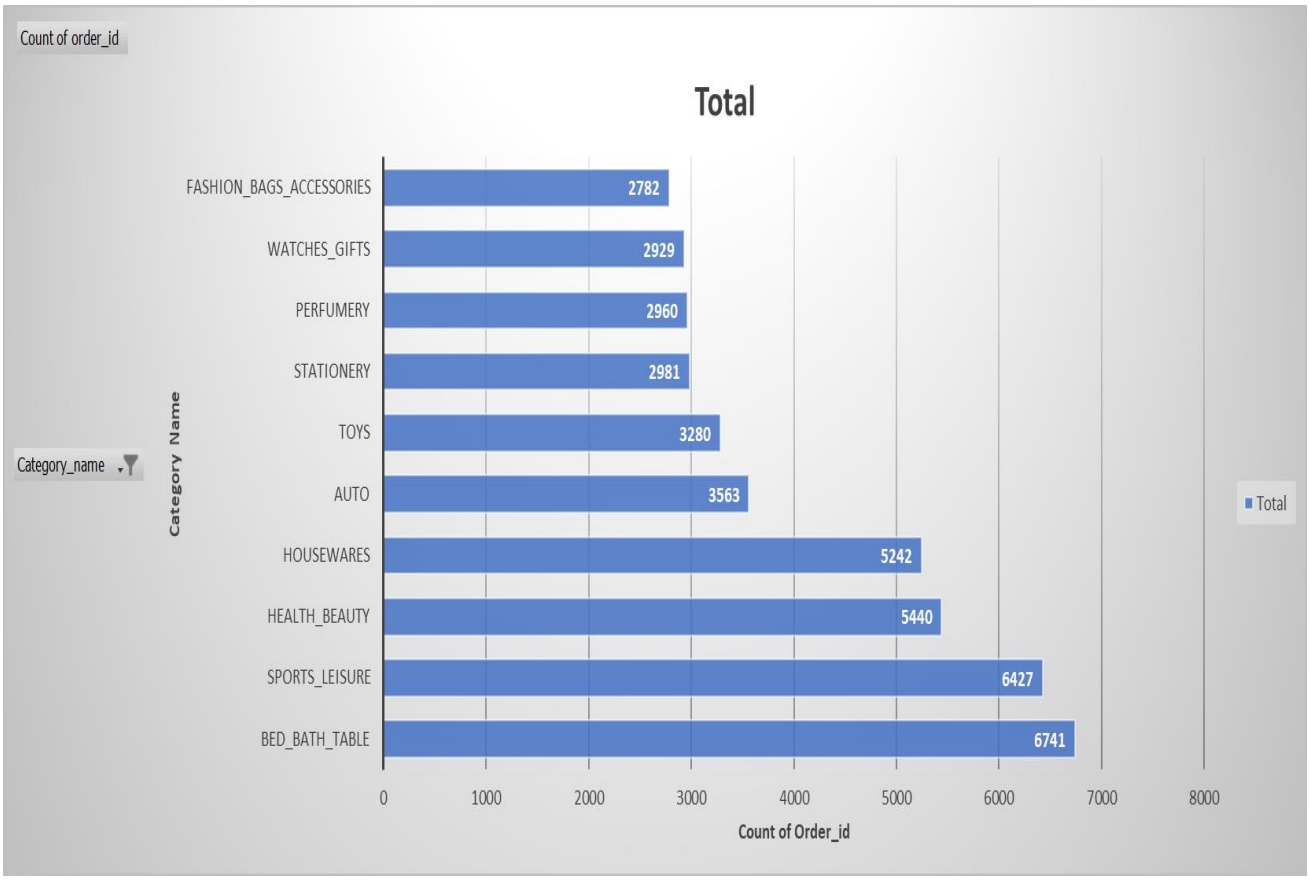


Business Insights:

The highest revenue-generating categories contribute a significant share of overall sales and should be prioritized for inventory planning, promotions, and marketing investments. Low-revenue categories may require pricing, assortment, or visibility improvements.

2. Which Categories Have the Top 10 Highest Order Volume?

Pivot Chart:

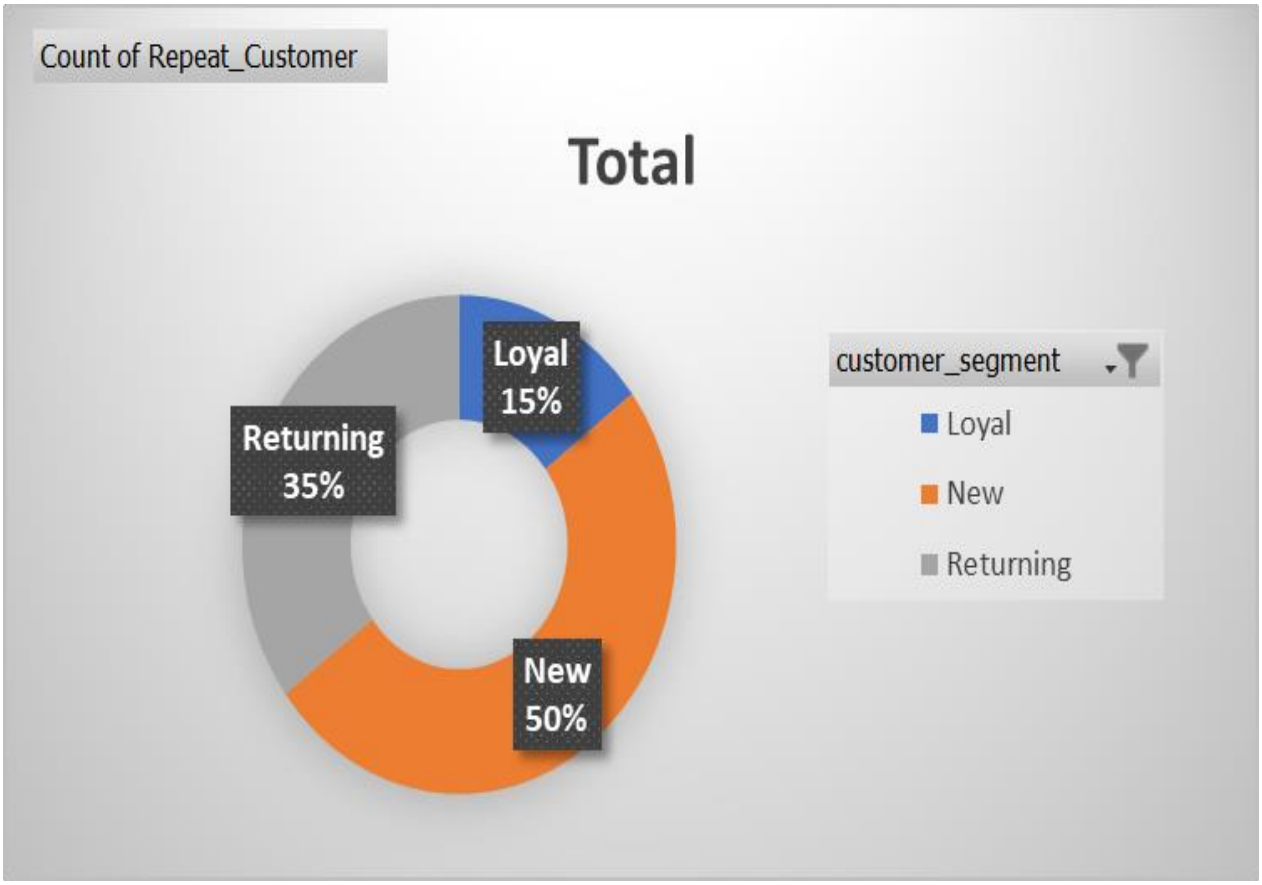


Business Insights:

High-order-volume categories indicate strong customer demand and purchasing frequency. These categories can be leveraged for cross-selling and customer acquisition strategies.

3. What is the Repeat Purchase Rate?

Pivot Chart:

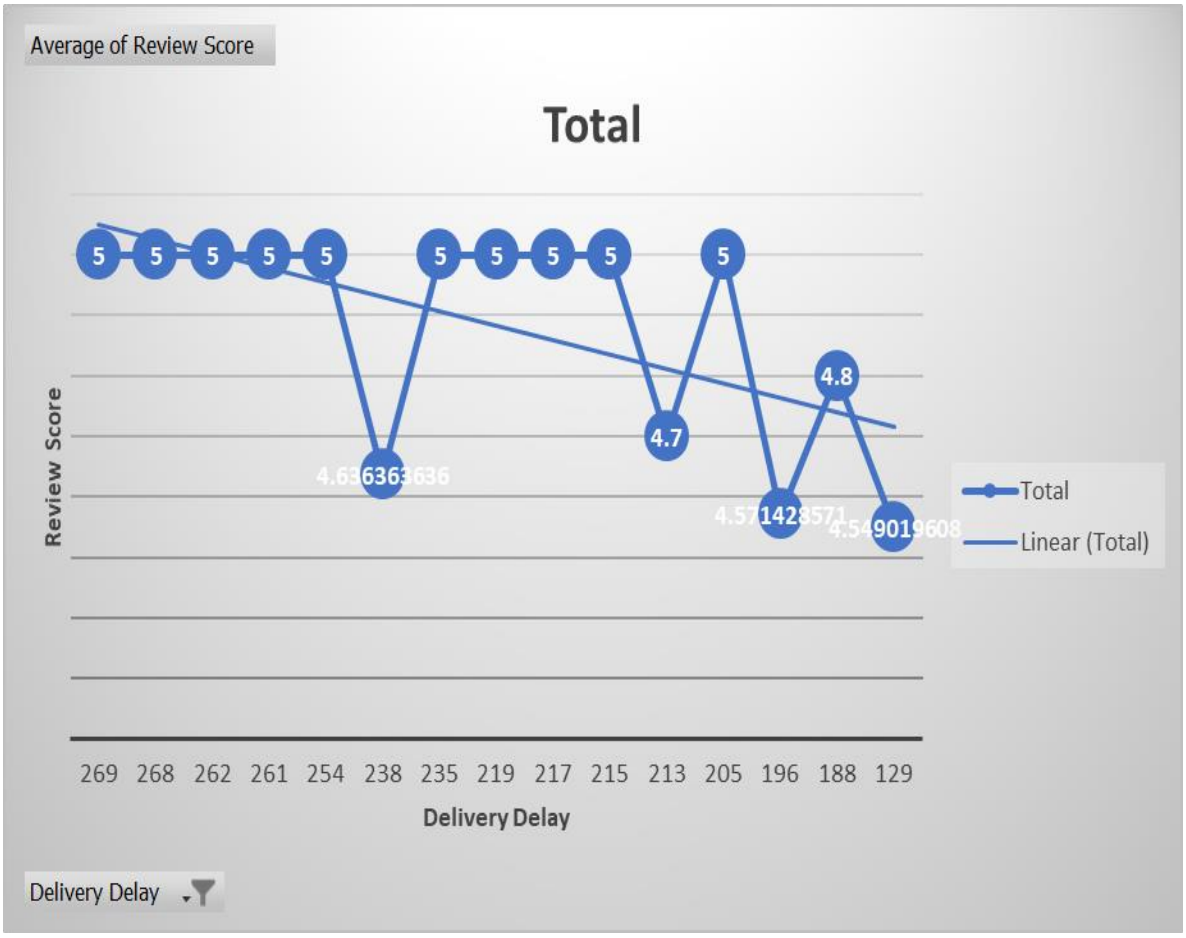


Business Insights:

A higher repeat purchase rate indicates strong customer loyalty and retention. If the proportion of new customers is high but repeat customers are low, retention strategies should be strengthened.

4. How Does Delivery Performance Affect Customer Satisfaction?

Pivot Chart:

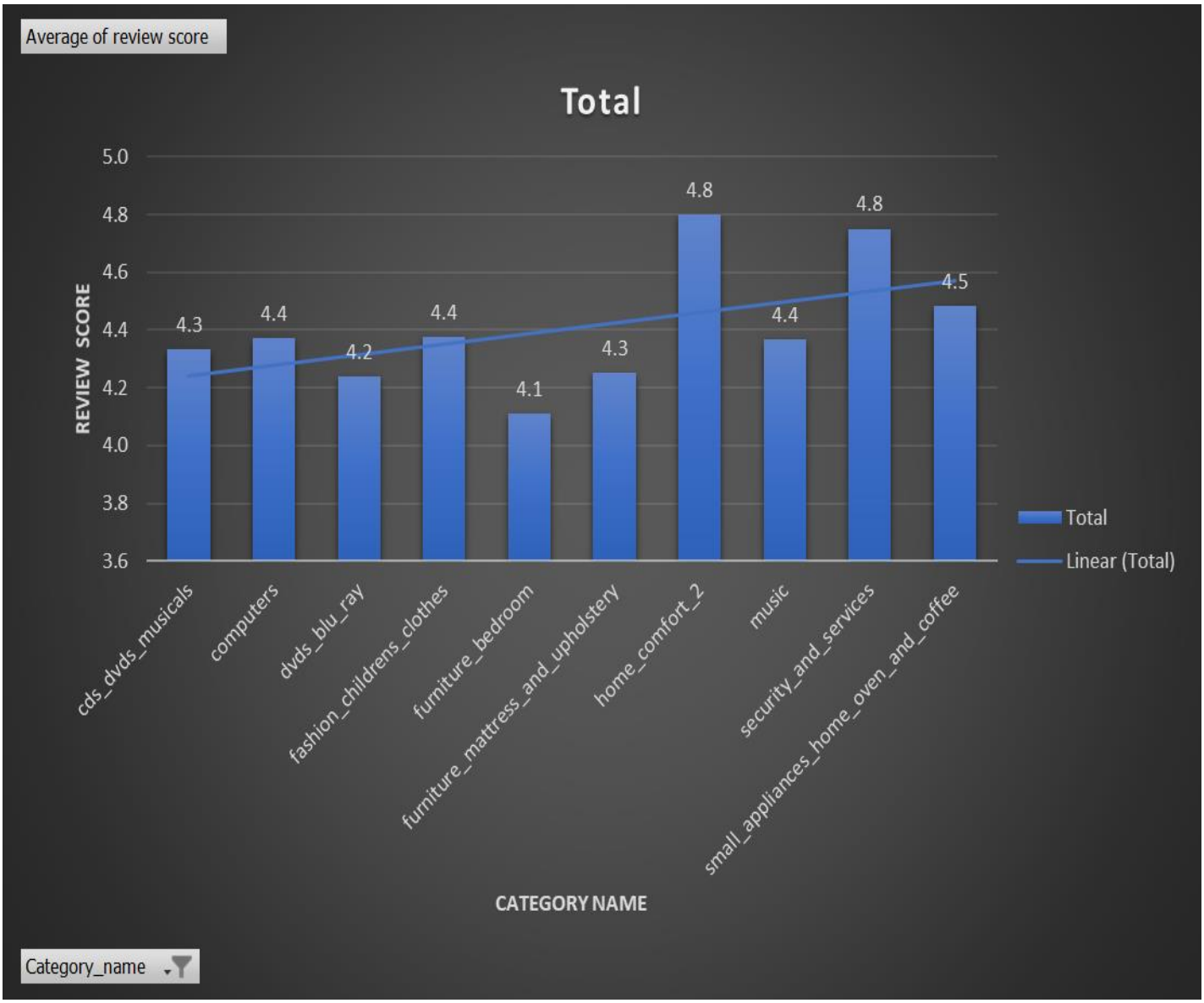


Business Insights:

Customers receiving orders on time generally provide better ratings, while delivery delays negatively impact satisfaction. Improving delivery efficiency can directly enhance customer experience and retention.

5. Which Product Categories Receive the Best Reviews?

Pivot Chart:

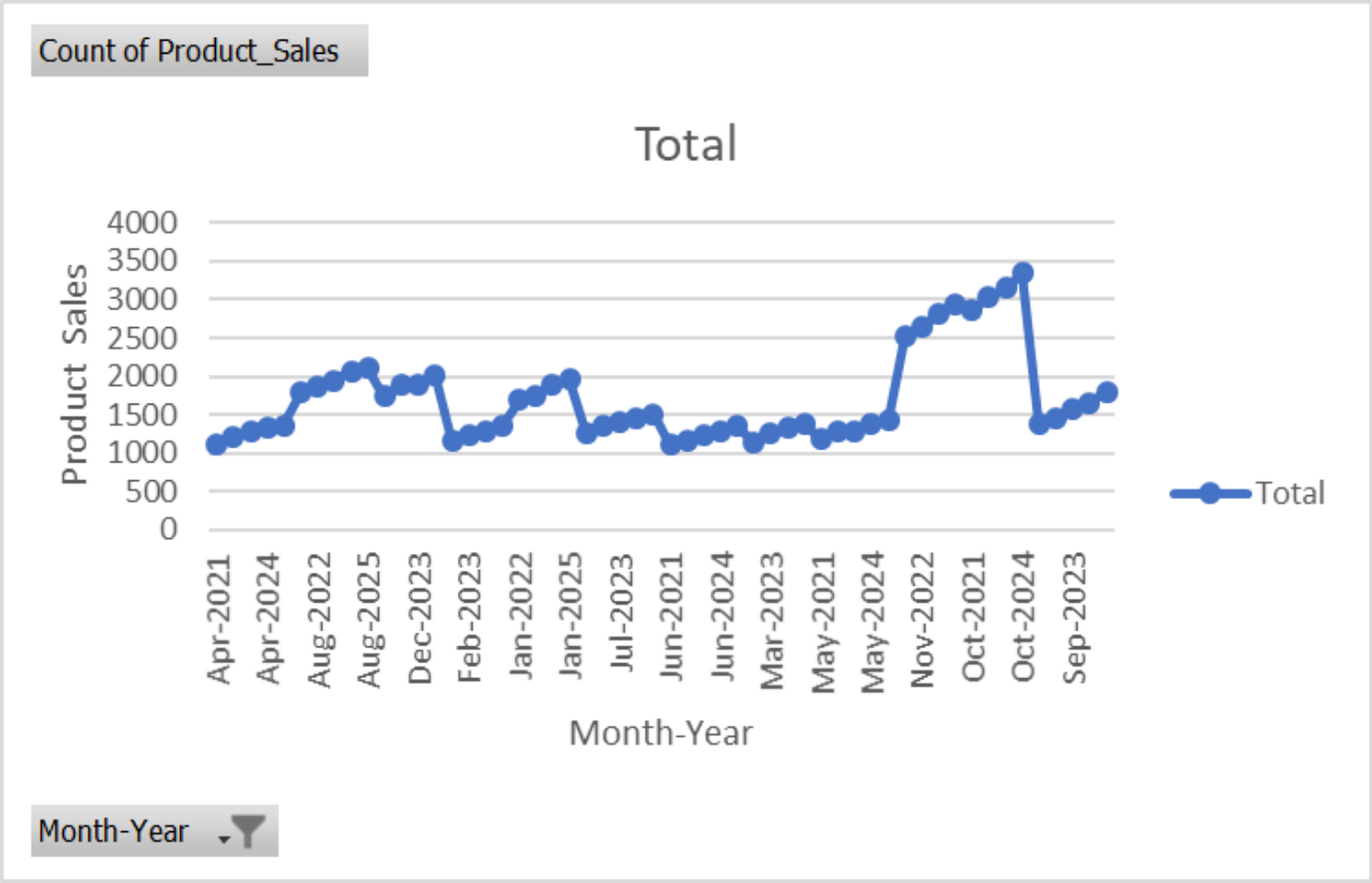


Business Insights:

Categories with higher average ratings demonstrate stronger customer satisfaction and product quality. These products can be promoted more aggressively to improve conversion rates.

6. How do sales revenues vary across different months?

Pivot Chart:

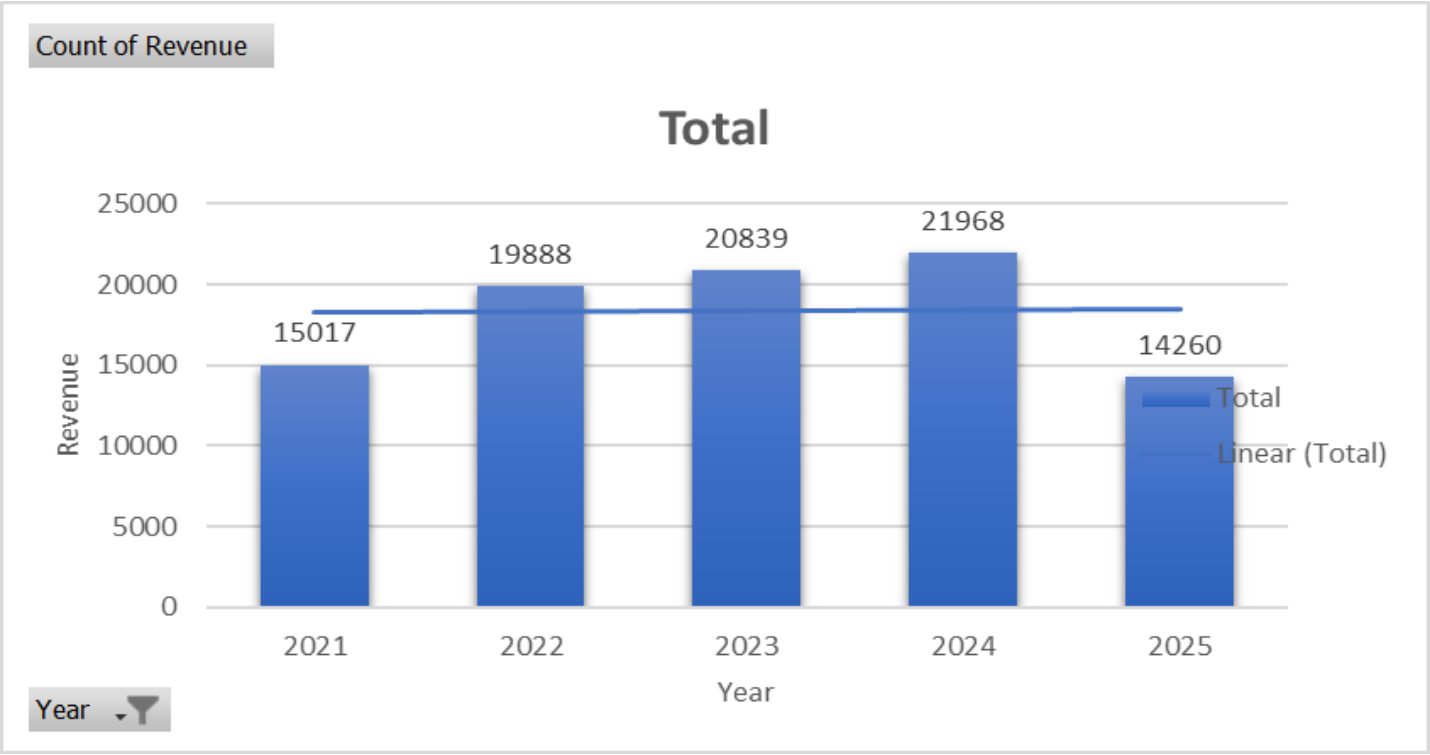


Business Insights:

Sales peaks and declines across months highlight seasonal buying behavior and demand patterns.

7. How has overall sales performance changed over the years?

Pivot Chart:

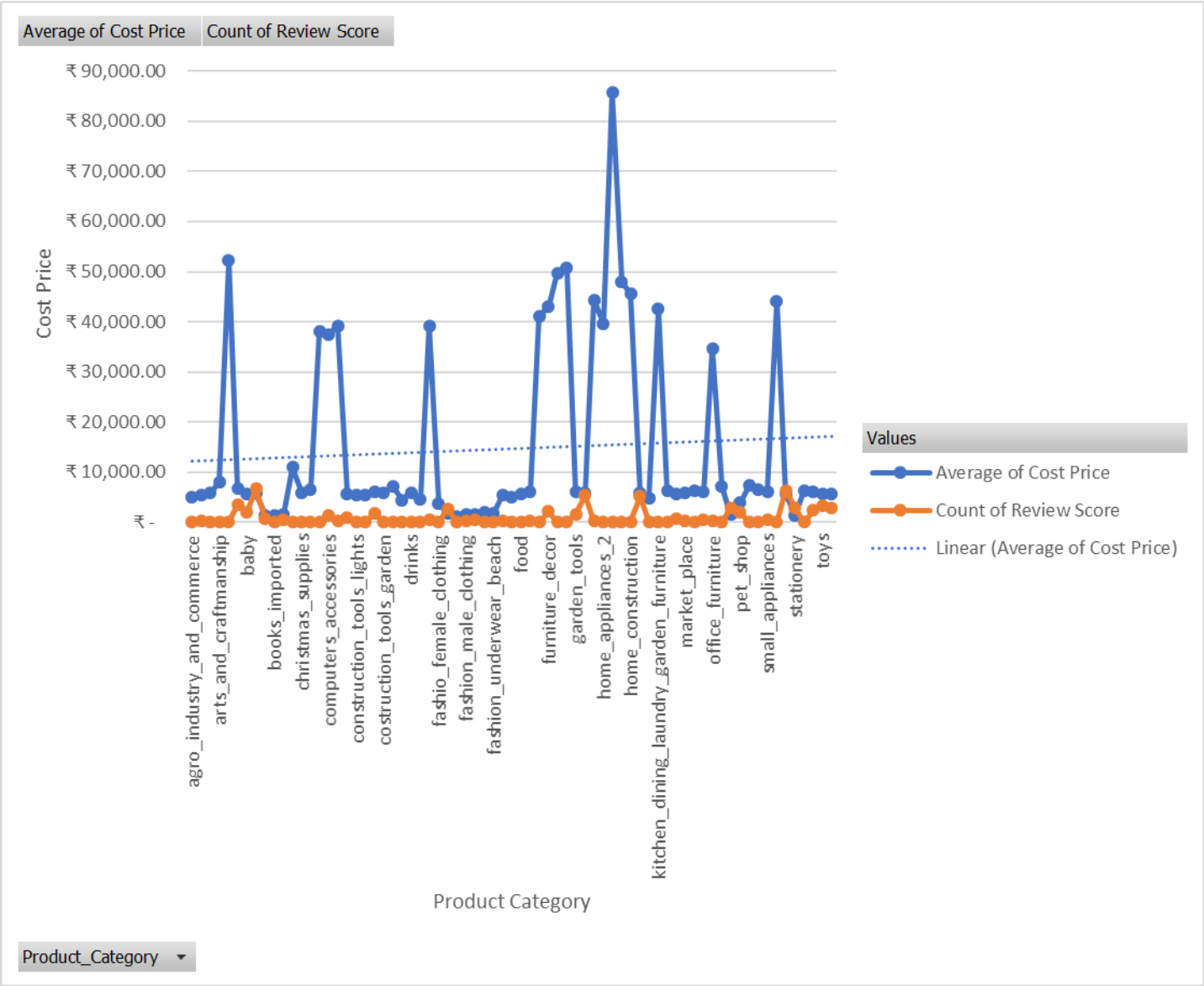


Business Insights:

Consistent year-over-year growth reflects increasing market penetration and customer demand.

8. How does product pricing influence customer satisfaction and review ratings?

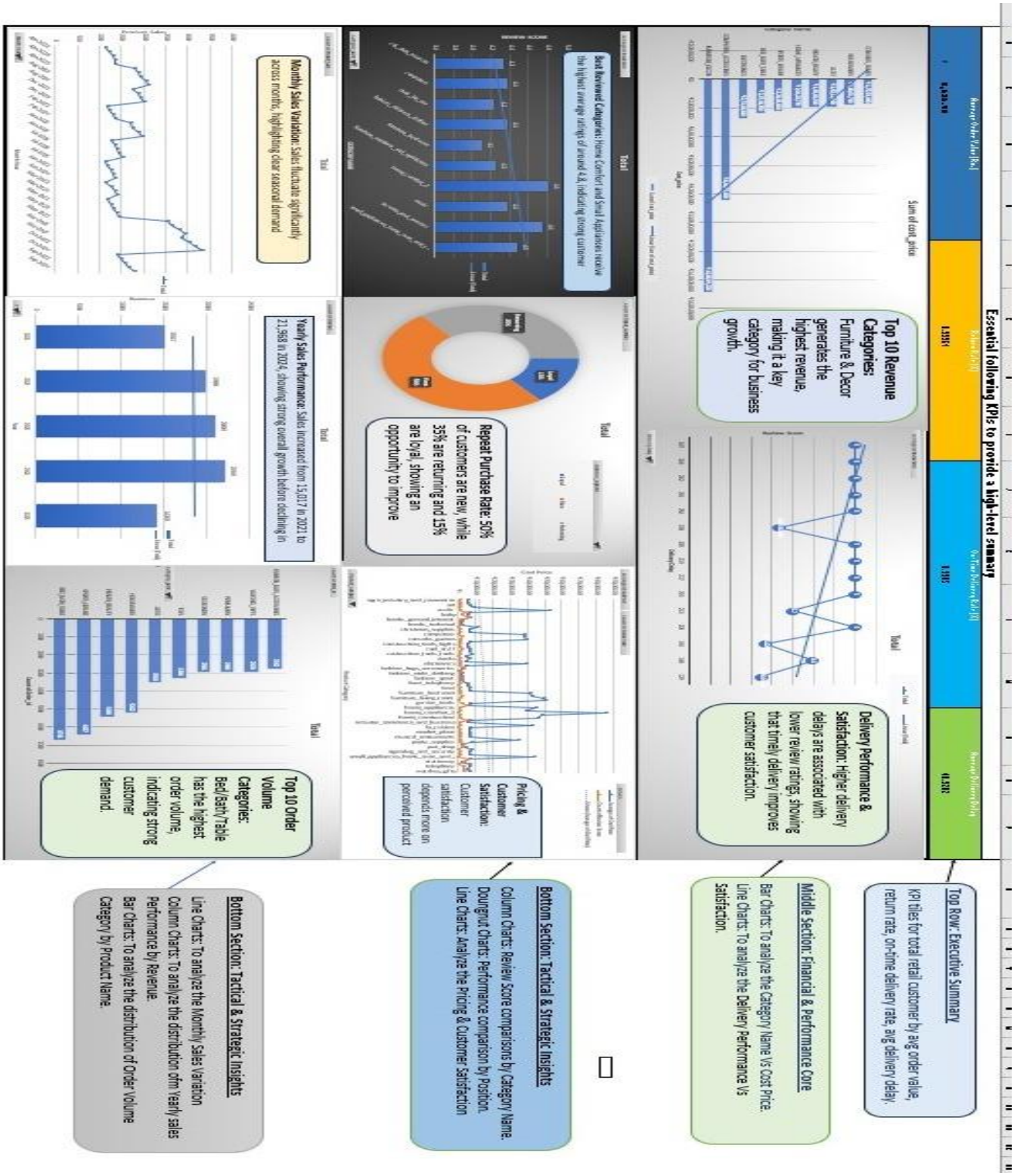
Pivot Chart:



Business Insights:

Customer ratings are influenced by perceived value rather than price alone, making quality a key driver of satisfaction.

Dashboard:


Essential following KPIs to provide a high-level summary
Top 10 Revenue Categories: Furniture & Decor generates the highest revenue, making it a key category for business growth.
Delivery Performance & Satisfaction: Higher delivery delays are associated with lower review ratings, showing that timely delivery improves customer satisfaction.
Best Reviewed Categories: Home Comfort and Small Appliances receive the highest average ratings of around 4.8, indicating strong customer
Repeat Purchase Rate: 50% of customers are new, while 35% are returning and 15% are loyal, showing an opportunity to improve
Pricing & Customer Satisfaction: Customer satisfaction depends more on perceived product
Monthly Sales Variation: Sales fluctuate significantly across months, highlighting clear seasonal demand
Yearly Sales Performance: Sales increased from 15,017 in 2021 to 21,968 in 2024, showing strong overall growth before declining in
Top 10 Order Volume Categories: Bed/Bath/Table has the highest order volume, indicating strong customer demand.
Top Row: Executive Summary
KPI tiles for total retail customer by avg order value, return rate, on-time delivery rate, avg delivery delay.
Middle Section: Financial & Performance Core
Bar Charts: To analyze the category Name Vs Cost Price.
Line Charts: To analyze the Delivery Performance Vs Satisfaction.
Bottom Section: Tactical & Strategic Insights
Column Charts: Review Score comparisons by Category Name.
Doungnut Charts: Performance comparison by Position.
Line Charts: Analyze the Pricing & Customer Satisfaction
Bottom Section: Tactical & Strategic insights
Line Charts: To analyze the Monthly Sales Variation
Column Charts: To analyze the distribution ofm Yearly sales Performance by Revenue.
Bar Charts: To analyze the distribution of Order Volume Category by Product Name.

Conclusion:

The dashboard analysis highlights the critical factors that influence the success of an e-commerce business, including product category performance, customer loyalty, delivery efficiency, and customer satisfaction. The findings indicate that a small number of product categories, customer segments, and geographic regions contribute significantly to overall revenue. Additionally, timely deliveries and positive customer experiences play an important role in maintaining high review scores and encouraging repeat purchases. By leveraging these insights, the business can optimize marketing strategies, strengthen customer retention initiatives, improve operational processes, and focus resources on high-performing products and markets to achieve long-term growth and profitability.